Remidify

Product Requirements Document and build brief

Fast swipe-card reminders with workspaces. iPhone-first web app, installed to the Home Screen, hosted on Vercel.

0. Brief for the AI coding agent (read first)

You are building **Remidify**, a single-user-first reminder web app as an installable PWA. The owner uses an **iPhone** and will add the site to the Home Screen. Follow this document as the source of truth. Where a choice is already made below, do not re-debate it; build it.

Non-negotiable goals

- **Speed:** a reminder can be captured in 3 taps or fewer and under 5 seconds.
- **Swipe cards:** the home screen is a Bumble-style stack of reminder cards with swipe actions.
- **Workspaces:** Main (shows everything) plus user-made ones such as Business and Job (show only their own reminders).
- **One-time login:** sign in once inside the installed app and stay signed in until Logout.
- **Reliable alerts:** Web Push notifications to the iPhone at the due time.
- **Apple Calendar:** a subscribable calendar feed (ICS) so reminders appear in Apple Calendar.
- **Mobile only design:** design at 390 px width first; desktop only needs to not break.

Locked technical decisions

Area	Decision
Framework	Next.js (App Router) + TypeScript, deployed on Vercel
Styling / UI	Tailwind CSS; Framer Motion for swipe physics; lucide-react icons
Database	Neon Postgres (Vercel integration) with Drizzle ORM
Auth	Auth.js (NextAuth v5). Google sign-in plus 6-digit email code. Database sessions, 365 day rolling maxAge
Push	Web Push with VAPID keys using the `web-push` package and a service worker
Scheduler	Endpoint `/api/cron/dispatch` called every minute by a free external scheduler (cron-job.org or Upstash QStash schedule), protected by a secret header. Vercel Hobby cron is daily only, so do not rely on it
Date parsing	chrono-node for natural language dates
PWA	Manifest + service worker (hand written or Serwist). Standalone display mode
Calendar	Per-user secret ICS feed URL at `/api/ics/[token].ics`

1. Problem and goals

Problem. Many tasks and follow-ups are being missed because recording them is slow and they live in too many places.

Goal. A very fast phone app to capture reminders, see only what matters now, and clear them with a swipe.

Success metrics

- Capture a reminder in under 5 seconds (3 taps or fewer).

- Opens in under 1 second from the Home Screen icon after first load.
- Push notification arrives within 60 seconds of due time.
- User is never asked to log in again until they tap Logout.

Out of scope for v1

- Team sharing, comments, attachments, native apps, desktop-optimised layouts.

2. Core concepts

Concept	Meaning
Reminder	Title, optional note, due date and time, workspace, status, optional repeat, priority.
Workspace	A container such as Main, Business, Job. Unlimited; each has name, colour and emoji.
Main workspace	Default home. Shows reminders from **all** workspaces, each with a small coloured tag. Cannot be deleted.
Workspace view	Opening Business shows **only** Business reminders. New reminders default to the workspace you are in.
Activity log	History of every change to a reminder. This is the 'log file with track' box in the original sketch.

3. Workflow

This follows the hand-drawn flow: **Remidify -> Set up reminder -> (Set due date / Same-day reminder / Log with track) -> Google or Apple Calendar integration if possible.**

3.1 First-time flow

1. Open the site in Safari, then Share > **Add to Home Screen**. Show a short guide screen explaining this if the app is not in standalone mode.
2. Open the app from the Home Screen icon. **Sign in inside the installed app** (see iOS notes in section 8).
3. Main workspace is created automatically.
4. Prompt: 'Enable notifications' (must be triggered by a button tap).
5. Land on the card stack.

3.2 Returning flow

1. Open the app, already signed in.
2. See today's and overdue cards for the current workspace.
3. Swipe to act, or tap + to capture.

3.3 Create reminder flow

1. Tap the + button (bottom centre, thumb reach). A bottom sheet opens and the keyboard focuses the title field.
2. Type the title. Date words are parsed live ('call vendor tomorrow 5pm'). The parsed date shows as an editable chip.
3. **Same day:** quick chips: In 1 hr, This evening (6 PM), Tonight (9 PM).
4. **Another date:** chips: Tomorrow 9 AM, Next Mon 9 AM, Pick date (native date and time input).
5. Optional: repeat (none, daily, weekly, monthly), workspace, priority. Hidden behind a 'More' toggle to keep the sheet fast.
6. Tap Save. Reminder is stored, a log entry is written, the card appears if due today, and the ICS feed includes it.

3.4 Reminder lifecycle

```
Created -> Scheduled -> Due (push sent) -> Done
                            |-> Snoozed (new due time) -> Due ...
                            |-> Rescheduled (new date)  -> Scheduled
                            |-> Deleted
Overdue items stay at the top of the stack until handled.
```

3.5 Workspace flow

1. Tap the workspace switcher in the top left.
2. Choose Main, Business, Job, or '+ New workspace' (name, colour, emoji).
3. Main shows everything with filter chips per workspace. Any other workspace shows only its own reminders.

3.6 Calendar flow

1. Settings > 'Add to Apple Calendar' opens a `webcal://` link. iOS shows a Subscribe prompt.
2. Per workspace toggle decides what appears in the feed.
3. Optional later: Google Calendar OAuth two-way sync.

4. Functional requirements

4.1 Authentication and access (P0)

- Google sign-in and email 6-digit one-time code. Do **not** use clickable magic links (see iOS notes).
- Session cookie: httpOnly, secure, SameSite=Lax, maxAge 365 days, rolling. User remains signed in until Logout.
- Logout button in Settings clears the session and unsubscribes that device's push endpoint.
- Every API route and query is scoped to the signed-in user id. Unauthenticated requests are redirected to /login.

4.2 Swipe card home (P0)

Gesture	Action
Swipe right	Mark done (green hint). Undo toast for 5 seconds.
Swipe left	Open snooze chips: 1 hr, Tonight, Tomorrow, Pick date.
Swipe up	Reschedule to another date (opens date picker sheet).
Tap card	Open detail and edit sheet.
Long press	Menu: move to workspace, delete.

- Card shows: title, due time (relative, e.g. 'Today 5:00 PM'), workspace tag, note preview, overdue badge.
- Stack ordering: overdue first, then due today by time, then upcoming. Show 'All clear' empty state with confetti-free minimal design.
- A toggle switches between Card view and List view. List view supports the same swipe actions on rows.
- Swipe threshold 35% of width; spring back if not passed. 60 fps; use transform and opacity only. Add haptic-like feedback via short vibration where supported.
- Optimistic updates: UI changes instantly, server sync in the background with retry.

4.3 Quick capture (P0)

- Floating + button always visible on home. Sheet opens in under 150 ms.
- Natural language dates via chrono-node ('tmrw 6pm', 'in 2 hours', 'next friday').
- If no date is parsed, default to 'In 1 hr' and show the chip so it can be changed.

- Enter key on the keyboard saves. Keyboard stays open for adding another reminder ('Save and add another' option).

4.4 Notifications (P0)

- Web Push to all registered devices at `due_at`. Notification actions are unreliable on iOS, so tapping the notification opens the app to that card.
- Dispatcher runs every minute: select reminders where status is open, `due_at <= now()` and `notified_at` is null. Send push, then set notified_at. Must be idempotent so no duplicates.
- Snooze or reschedule resets notified_at to null.
- Repeat rules: when a repeating reminder is marked done, create the next occurrence.
- Daily digest push at the user's chosen time (default 8:00 AM) listing the day's reminders, per user timezone.
- Remove push subscriptions that return HTTP 404 or 410.

4.5 Workspaces (P0)

- Create, rename, recolour, archive, delete (delete asks what to do with its reminders: move to Main or delete).
- Switcher remembers the last used workspace.
- Strict filtering: queries for a workspace view always include the workspace id. Main omits the filter.

4.6 Calendar integration (P1)

- ICS feed: VCALENDAR with one VEVENT per open reminder in synced workspaces (30 minute duration, VALARM at time zero), stable UID per reminder.
- Include `REFRESH-INTERVAL;VALUE=DURATION:PT15M` and `X-PUBLISHED-TTL:PT15M`. Apple controls the real refresh timing, so the calendar can lag. Push notifications are the reliable alert.
- Feed token is a long random string, regenerable in Settings.
- Per reminder 'Add to Calendar' downloads a single .ics file.
- Apple does not provide a web API to write to the calendar directly, so the subscribed feed is the supported method. Google Calendar API sync is P2.

4.7 Activity log (P1)

- Record: created, edited, snoozed, rescheduled, completed, reopened, deleted, with old and new values.
- Reminder detail shows its history. A global History screen filters by workspace and date.

4.8 Data and offline (P0/P1)

- All data lives in Postgres. Reads use cached data first (stale-while-revalidate) so the app opens instantly.
- Offline: view cached reminders, create and complete while offline, queue changes in IndexedDB and sync on reconnect (P1).
- Export all data as JSON (P2).

5. Screens and UI rules

Screen	Contents
Login	Logo, 'Continue with Google', email field, 6-digit code entry.
Home	Top: workspace switcher, view toggle. Centre: card stack. Bottom: tab bar (Home, History, Settings) with centred + button.
Quick add sheet	Title input, date chips, 'More' for repeat, workspace, priority. Save button above keyboard.
Reminder detail	Edit fields, history list, delete.

Screen	Contents
Workspaces	List with counts, create, edit, archive.
History	Global activity log with filters.
Settings	Notifications on/off and test button, digest time, timezone, calendar feed link, logout.

- Layout: 390 px wide first. Use `100dvh`, safe-area insets (`env(safe-area-inset-*)`) and viewport-fit=cover.
- Primary actions live in the bottom third of the screen (thumb zone). Minimum tap target 44 px.
- Use `touch-action: pan-y` on cards and prevent pull-to-refresh and overscroll conflicts. Inputs use font-size 16 px or more to stop iOS zoom.
- Dark and light mode follow the system. Clean, minimal, large type. Workspace colour used as an accent on cards.
- No heavy libraries. Keep the first-load JS small.

6. Data model

```
users            id, email, name, image, timezone, digest_time, calendar_token, created_at
sessions / accounts / verification_tokens   (Auth.js tables)
workspaces       id, user_id, name, colour, emoji, is_main, archived, sync_to_calendar,
                 sort_order, created_at
reminders        id, user_id, workspace_id, title, note, due_at (timestamptz), status
                 ('open'|'done'), priority (0-2), repeat_rule (null|'daily'|'weekly'|'monthly'),
                 notified_at, snoozed_count, completed_at, created_at, updated_at
activity_log     id, reminder_id, user_id, action, old_value (jsonb), new_value (jsonb), created_at
push_subs        id, user_id, endpoint (unique), p256dh, auth, user_agent, created_at

Indexes: reminders(user_id, workspace_id, status, due_at); reminders(status, notified_at, due_at)
Rules: store all times in UTC; render in users.timezone.
```

7. API routes

Route	Purpose
GET/POST /api/reminders	List (filters: workspace, status, range) and create
PATCH/DELETE /api/reminders/[id]	Edit, done, snooze, reschedule, delete (all write activity_log)
GET/POST /api/workspaces, PATCH/DELETE /api/workspaces/[id]	Workspace CRUD
POST /api/push/subscribe, POST /api/push/unsubscribe	Store or remove a push subscription
GET /api/cron/dispatch	Send due notifications. Requires header x-cron-secret
GET /api/ics/[token].ics	Calendar feed
GET /api/ics/reminder/[id].ics	Single reminder file
GET /api/history	Activity log

8. iPhone and PWA requirements (critical)

- **Add to Home Screen is required for push.** iOS Web Push (iOS 16.4+) only works when the site is installed. Detect standalone mode with `navigator.standalone` or `display-mode: standalone`; if not installed, show install instructions instead of the notification prompt.
- **Separate storage.** Safari and the installed app do not share cookies. The user must sign in inside the installed app. Hence the 6-digit code or Google sign-in. Clickable email magic links open Safari, not the app,

so avoid them.

- **Permission prompt** for notifications must follow a user tap (button in onboarding and Settings).
- Manifest: name 'Remidify', short_name 'Remidify', display 'standalone', start_url '/', theme and background colours, icons 192, 512 and maskable. Also add `apple-touch-icon` (180 px) and `apple-mobile-web-app-capable` meta tags.
- Service worker handles `push` and `notificationclick` (focus or open the app at the reminder).
- Notification action buttons are not reliable on iOS, so do not depend on them.
- Calendar subscribe link (webcal://) should open in Safari via a normal link tap.

9. Non-functional requirements

- **Performance:** Lighthouse mobile performance 90+, interaction under 100 ms, optimistic UI.
- **Security:** HTTPS, httpOnly cookies, validate input with zod, per-user ownership checks on every query, cron endpoint secret, regenerable calendar token, no secrets in client code.
- **Reliability:** idempotent dispatch, retry on push failure, no duplicate notifications.
- **Time zones:** UTC storage, per-user timezone display. Handle DST correctly for repeating reminders.
- **Accessibility:** labelled buttons, contrast AA, support Dynamic Type via relative units.

10. Environment variables (Vercel)

```
DATABASE_URL=            # Neon Postgres connection string
AUTH_SECRET=             # random 32+ chars
AUTH_GOOGLE_ID=  AUTH_GOOGLE_SECRET=
EMAIL_SERVER / RESEND_API_KEY=   # to send the 6-digit login code
VAPID_PUBLIC_KEY=  VAPID_PRIVATE_KEY=  NEXT_PUBLIC_VAPID_PUBLIC_KEY=
VAPID_SUBJECT=mailto:you@example.com
CRON_SECRET=             # checked against x-cron-secret
NEXT_PUBLIC_APP_URL=https://your-app.vercel.app
```

11. Build plan for the agent

Build in this order. Finish and verify each phase before the next. Commit after each phase.

Phase	Deliverables	Done when
1. Foundation	Next.js + Tailwind + Drizzle + Neon schema, PWA manifest, icons, layout with safe areas.	App installs to iPhone Home Screen and opens full screen.
2. Auth	Google sign-in, email code, 365 day session, protected routes, logout.	Sign in once in the installed app; still signed in after days.
3. Reminders core	CRUD API, quick-add sheet with chrono-node, date chips, list view.	Reminder created in 3 taps; same-day and future dates work.
4. Swipe cards	Card stack, gestures (done, snooze, reschedule), undo, empty state.	All gestures work smoothly on a real iPhone.
5. Workspaces	Switcher, create/edit/archive, Main aggregate, strict filtering.	Business view shows only Business reminders.
6. Push	Service worker, subscribe flow, dispatch endpoint, scheduler setup, digest.	Notification arrives within a minute of due time.
7. Calendar and log	ICS feed, per-reminder .ics, activity log and History screen.	Apple Calendar subscription shows reminders.
8. Polish	Offline queue, repeat rules, performance pass, dark mode, empty/error states.	Lighthouse 90+, offline create works.

12. Acceptance criteria

1. After signing in once inside the Home Screen app, closing and reopening days later still shows me signed in until I tap Logout.
2. I can add a reminder for later today in 3 taps or fewer, and for another date with the date chips or picker.
3. A reminder for a future date does not appear in today's stack until that day (it appears in upcoming/list view).
4. Opening Business shows only Business reminders; Main shows all with workspace tags.
5. Swipe right completes, swipe left offers snooze, swipe up reschedules, and undo restores.
6. A push notification arrives on my iPhone at the due time and tapping it opens the reminder.
7. Subscribing to the feed in Apple Calendar shows my open reminders from synced workspaces.
8. Every change appears in the activity log.
9. The app works at 390 px width with no horizontal scroll, and respects the notch and home indicator.

13. Assumptions and open items

- Assumed: Main shows reminders from all workspaces. If you want Main to contain only its own reminders, change the Main query to filter by its workspace id.
- Assumed: single user now; schema already supports more users.
- Open: add Google Calendar two-way sync later? Quiet hours per workspace? Repeat rules in the first release?
- Risk: Apple Calendar feed refresh is controlled by iOS and may lag. Mitigation: push notifications are the primary alert.

14. Starter prompt to paste into the agent

```
Read the attached Remidify PRD fully. Build the app phase by phase as described in
section 11, starting with Phase 1. Use exactly the locked technical decisions in
section 0. Target iPhone Home Screen PWA, 390 px mobile-first. After each phase, run
the app, summarise what works, list anything unverified (especially push and iOS
behaviour that needs a real device), and wait for my confirmation before starting
the next phase. Ask me for any API keys or accounts you need instead of inventing them.
```